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An efficient lung sound separation algorithm base on GIHO-VMD

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ABSTRACT

Background and Objective: Lung sounds (LS) serve as a critical source of pathological information for diagnosing lung diseases. However, during clinical auscultation, LS are often contaminated by heart sounds (HS) and other noises, significantly compromising the accuracy of pathological assessment. The primary challenge lies in the spectral overlap between LS and HS, coupled with the non-stationary nature of both signals. Furthermore, abnormal LS exhibit intricate frequency components and ambiguous band boundaries, causing conventional separation methods to yield incomplete or contaminated results, which hinders precise diagnosis.

Methods: To overcome these challenges, we propose a Guided-Improve Hippopotamus Optimization based Variational Mode Decomposition (GIHO-VMD) algorithm for LS separation. Specifically, a VMD model is first established with Fuzzy Entropy (FE) as the objective function. Critically, a novel complexity constraint is incorporated into this model to effectively restrict decomposition outcomes. The Hippopotamus Optimization (HO) algorithm is then employed to iteratively optimize the VMD parameters based on this objective function. To prevent the optimization from converging to local optima, a Disruption Operator (DO) is integrated into HO. Subsequently, a Guided Learning Strategy (GLS) is introduced to enhance the guiding influence of superior agents during iteration, thereby improving convergence accuracy. Finally, a dual-process dynamic threshold mechanism is implemented within GLS to moderate the frequency of guidance, thereby stabilizing the exploration–exploitation balance throughout the optimization process.

Results: To evaluate the proposed method, we apply it to separate pre-simulated mixed signals of HS and LS. The SNR and NCC between the separated LS and the original ones are evaluated. Experimental results indicate that the method achieves an average SNR of 30.27 dB for normal lung sounds and an average SNR ranging from 28.60 dB to 30.11 dB for abnormal lung sounds, with an average NCC of 63.15%.

Conclusion: Our method demonstrates robust capabilities in lung sound signal separation, effectively addressing the challenge of extracting LS from heart and lung sound(HLS), and shows significant potential for enhancing the generalization, quality, and accuracy of lung sound signal decomposition.

1. Introduction

Lung sounds (LS) are crucial for medical diagnosis as they facilitate the identification of pulmonary pathologies. However, gastrointestinal and muscle friction noises can significantly degrade the accuracy of LS auscultation [1]. Furthermore, heart sounds (HS) may spectrally overlap with LS in clinical recordings, leading to aliasing interference [2], which can significantly compromise the performance of both Discrete Wavelet Transform (DWT) and Empirical Wavelet Transform (EWT) in signal processing [3]—even though EWT holds certain advantages in handling weak signals [4,5]. To separate the LS from other noises, Empirical Mode Decomposition (EMD) has been widely applied to decompose mixed signals into intrinsic mode functions (IMFs) [6,7]. EMD can adaptively capture the time–frequency features of LS. However, its susceptibility to mode mixing and noise sensitivity often leads to ambiguous decomposition, particularly when HS and LS overlap spectrally [8,9]. Unlike EMD's heuristic sifting process, Variational Mode Decomposition (VMD) constructs a mathematically rigorous optimization framework. VMD [10] is utilized to optimize IMF bandwidth through a variational framework, effectively suppressing noise and resolving mode aliasing [11]. While the VMD algorithm employs an Alternating Direction Method of Multipliers (ADMM) to solve the variational problem, practical applications in LS separation often reveal limitations in sensitivity to the number of modes and the penalty parameters.

Fortunately, Metaheuristic Algorithms (MA) can be effectively employed to devise optimal parameter selection strategies for VMD [12]. By leveraging stochastic search mechanisms, the application of MAs

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greatly decreases the need for manual parameter tuning in VMD [13–15]. Unlike adaptive methods that iteratively refine parameters based on local signal characteristics, VMD is allowed to achieve more accurate decomposition by finding globally optimal parameter combinations through the search mechanisms of MAs. Moreover, the application of MA reduces VMD's sensitivity to initial parameters by exploring various parameter regions, thus enhancing the robustness of decomposition results [15,16].

Among MAs, Hippopotamus Optimization (HO) Algorithm is a novel MA designed by observing the inherent behaviors of hippopotamus groups, exhibiting high search capability and fast convergence speed [17]. Despite its advantages, HO shares a common MAs limitation: convergence to local optima during the exploitation phase, with limited capability to escape suboptimal solutions. To address this challenge, good point sets and chaotic mapping are used as initialization method in MAs to mitigate the impact of uneven initialization on the search [18,19]. However, the initialization phase has a limited impact on the entire algorithm and chaotic mapping depends on the combination with other improvement methods [20]. To this end, Opposition-Based Learning (OBL) strategies are developed to evaluate inverse solution sets [21]. While this approach enhances exploration diversity, it inadvertently diminishes the guidance toward high-quality solutions. Consequently, the Guided Learning Strategy (GLS) is introduced to dynamically assess the algorithm's state, thereby achieving a stable equilibrium between exploration and exploitation [22]. On the other hand, to minimize the risk of falling into local optima, Lévy flight strategy is employed to enhance the global search capability of the optimization algorithm [23,24]. Although Lévy flight strategy can perturb the optimal position to prevent algorithm converging to local optima, its parameter sensitivity limits improvement in the algorithm's convergence speed and accuracy. Furthermore, Disruption Operator (DO) is incorporated to define repulsion forces between candidate solutions [25]. This mechanism enhances global search capability while effectively preventing premature convergence [26,27].

To further strengthen the search capability during the iterative optimization process, this paper integrates HO with Guided Learning Strategy (GLS), yielding the Guided Improved Hippopotamus Optimization (GIHO) algorithm and introduces DO to prevent the algorithm from converging to local optima. Additionally, by adopting Fuzzy Entropy as the objective function, a GIHO-based Variational Mode Decomposition (GIHO-VMD) algorithm is proposed. The primary contributions of this study are summarized as follows:

(1) To strengthen the stability and improve convergence speed, a guided improvement strategy with dynamic threshold is introduced in exploitation to generate GIHO.
(2) Constructing Disruption Operator using the distance information of candidate solutions to enhance search capability of GIHO.
(3) An algorithm GIHO-based VMD(GIHO-VMD) with minimizing the intricacy of signal decomposition is proposed for lung sound separation.

2. Preliminaries

2.1. Variational Mode Decomposition

Variational Mode Decomposition is an adaptive, non-recursive signal processing technique. Nowadays a variety of modified versions of VMD have been developed. Research have demonstrated that although improved methods such as IVMD exhibit better performance in their respective enhanced aspects, they also compromise to some extent the general applicability of the original VMD. Therefore, the VMD algorithm employed in our study is the classical one proposed by Konstantin Dragomiretskiy and Dominique Zosso [10], which is described as follows:

$$\min_{\{u_k\},\{\omega_k\}} \left\{ \sum_k \left\| \partial_t \left[\left(\delta(t) + \frac{j}{\pi t} \right) * u_k(t) \right] exp(-j\omega_k t) \right\|_2^2 \right\} \quad s.t. \quad \sum_k u_k = f, \tag{1}$$

where u_k is set of all modes, ω_k is respective central frequencies of all modes. The summation over all modes is indicated by $\sum_k$ and f represents original signal in variational problem. The derivation of the solution for u_k and ω_k using Lagrangian multipliers with a quadratic penalty term is provided in Appendix.

2.2. Fuzzy Entropy

Fuzzy Entropy (FE) is used to assess the complexity of a signal. It examines the resemblance among sequences within a permissible range. Unlike conventional entropy calculations FE employs a fuzzy membership function [28] to enable a more refined evaluation of the pattern connections within the signal.

For a given signal sequence $\{s(i) : 1 \le i \le N\}$ and embedding dimension S_i^m, define the vector as

$$S_i^m = \{s(i), s(i+1), \dots, s(i+m-1)\} - s_0(i), \tag{2}$$

where $s_0(i) = \frac{1}{m}\sum_{j=0}^{m-1} s(i+j)$ is baseline correction term. The distance between vectors S_i^m and S_j^m is given by

$$d_{ij}^m = \max_{k\in(0,m-1)} |(s(i+k) - s_0(i)) - (s(j+k) - s_0(j))|. \tag{3}$$

Function $\exp(-(d_{\mathrm{ij}}^m)^n/r)$ is chosen as the fuzzy function, leading to

$$\phi^m(n,r) = \frac{1}{N-m}\sum_{i=1}^{N-m}\left(\frac{\sum_{j=1,j\neq i}^{N-m} \exp(\frac{-(d_{\mathrm{ij}}^m)^n}{r})}{N-m-1} \right), \tag{4}$$

where r is similarity tolerance, n regulates the sensitivity to the complexity of the time series. Finally, the Fuzzy Entropy is defined as

$$\mathrm{FuzzyEn}(s(i), m) = \ln \phi^m(n,r) - \ln \phi^{m+1}(n,r), \tag{5}$$

which serves as the objective function to be minimized in the optimization.

2.3. Hippopotamus optimization algorithm

HO is designed by observing the inherent behaviors of hippopotamus groups [17]. It is implemented through a three-phase model which includes updating the position of hippopotamus in rivers or ponds, their defensive strategies against predators, and their evasion methods to avoid predators. In this paper, HO is applied to compute the optimization problem with objective function $Min[FuzzyEn]$ to ensure that the decomposed signals have minimal complexity.

At the beginning of HO [17], the position of i th hippopotamus χ_i are initialized randomly within the search spacerange:

$$\chi_i : x_{i,j} = lb_j + r_1.\left(ub_j - lb_j\right), \tag{6}$$

where ub_j, lb_j are the lower and upper bounds of j th decision variable, $\mathcal{N}$ is the population size of hippopotamuses within the herd and m denotes the number of decision variables in the problem, with $1 \le i \le \mathcal{N}$, $1 \le j \le m$. Let $\chi_i^{\mathcal{M}hip}$ be male hippopotamuses, which can be defined as

$$\chi_i^{\mathcal{M}hip} : x_{i,j}^{\mathcal{M}hip} = x_{i,j} + r_2.\left(\mathcal{D}h - I_1 x_{i,j}\right), \tag{7}$$

where $\mathcal{D}h$ is the individual with the highest score, $I_1, I_2 \in \{1,2\}$. Position of female or immature hippopotamuses χ_i^{FBhip} can be calculated by

$$x_{i,j}^{\mathcal{FB}hip} = \begin{cases} x_{i,j} + h_1 \cdot (\mathcal{D}h - .I_2\mathcal{M}_i) & T > 0.6 \\ x_{i,j} + h_2 \cdot (\mathcal{M}_i - \mathcal{D}h) & T \le 0.6 \& r_3 > 0.5 \\ lb_j + r_4 \cdot \left(ub_j - lb_j\right) & else, \end{cases} \tag{8}$$

$$h = \begin{cases} I_2 \times \vec{r}_5 + (\sim \varrho_1) \\ 2 \times \vec{r}_6 - 1 \\ \vec{r}_7 \\ I_1 \times \vec{r}_8 + (\sim \varrho_2) \\ r_9, \end{cases} \tag{9}$$

here ϱ_1 and ϱ_2 are integer random numbers which can be one or zero, $r_{1,\ldots,4} \in [0,1]$, $\vec{r}_{5,\ldots,11} \in \mathbb{R}^{1\times\mathcal{N}}$. T is defined as $T = \exp(-t/\mathcal{T})$, with t and $\mathcal{T}$ representing the current iteration number and the maximum number of iterations, respectively.

Let $\mathcal{P}_j$ be the predator's position in search space. It can be calculated as

$$\mathcal{P} : \mathcal{P}_j = lb_j + \vec{r}_{10} \cdot \left(ub_j - lb_j\right), j = 1, 2, \ldots, m. \tag{10}$$

Thereafter, the process of hippopotamuses approaching predators is described as (11) with $\overline{D} = \left|\mathcal{P}_j - x_{i,j}\right|$ denoting the distance of the ith hippopotamus to the predator.

$$\chi_i^{\mathcal{H}i} : x_{i,j}^{\mathcal{H}i} = \begin{cases} \overrightarrow{RL} \oplus \mathcal{P}_j + \mathcal{R}a.\left(\frac{1}{\overline{D}}\right) & \mathcal{F}_{\mathcal{P}r_j} < \mathcal{F}_i \\ \overrightarrow{RL} \oplus \mathcal{P}_j + \mathcal{R}a.\left(\frac{1}{2\times\overrightarrow{D}+\vec{r}_{11}}\right) & \mathcal{F}_{\mathcal{P}r_j} \ge \mathcal{F}_i \end{cases} \tag{11}$$

$$\mathcal{R}a = \left(\frac{f}{(c - d \times cos(2\pi g))}\right). \tag{12}$$

As defined, $\mathcal{F}_i$, $\mathcal{F}_{\mathcal{P}r_j}$ is objective function value of $x_{i,j}^{\mathcal{H}i}$ and $\mathcal{P}_j$. $\chi_i^{\mathcal{H}i}$ represents the position of the hippopotamus facing predator. $\mathcal{R}a$ is random weight called displacement distance factor with $f \in [2,4]$, $c \in [1, 1.5]$, $d \in [2,3]$, $g \in [-1,1]$. $\overrightarrow{RL}$ is a random vector with a Levy distribution, which is calculated as

$$\mathcal{L}evy(\vartheta) = 0.05 \times \left[\frac{\Gamma(1+\vartheta)\sin\left(\frac{\pi\vartheta}{2}\right)}{\Gamma\left(\frac{(1+\vartheta)}{2}\right)\vartheta 2^{\frac{(\vartheta-1)}{2}}}\right]^{\frac{1}{\vartheta}} \times \frac{w}{|v|^{\frac{1}{\vartheta}}}, \tag{13}$$

with $\Gamma(\cdot)$ being the Gamma function, $w, \sigma \in [0,1]$, $\vartheta = 1.5$.

Next, update boundaries of the current phase as (14), here, ub_j^{cur} and lb_j^{cur} represent the current upper and lower bounds of the search space, respectively.

$$lb_j^{cur} = \frac{lb_j}{t}, ub_j^{cur} = \frac{ub_j}{t}, t = 1, 2, \ldots, \mathcal{T} \tag{14}$$

Then, applying the boundary in (14) yields a new hippopotamus population as

$$\chi_i^{\mathcal{H}ip\mathcal{E}} : x_{i,j}^{\mathcal{H}ip\mathcal{E}} = x_{i,j} + r_{12} \cdot \left(lb_j^{cur} + s \cdot \left(ub_j^{cur} - lb_j^{cur}\right)\right), \tag{15}$$

with $s \in [-1,1]$ and $r_{12} \in [0,1]$.

After completing each iteration of the HO algorithm, all population members are updated based on Eqs. (7)–(15). The process continues until the final iteration.

3. Method

3.1. Guided-improved HO

In order to enhance the search ability of algorithm and stabilize the exploration–exploitation balance during the optimization process, this paper introduces Guided Learning Strategy (GLS) [22] and Disruption Operator (DO) [25] to improve HO, referred to as GIHO.

GLS employs experience accumulation to discern the state of an algorithm by analyzing its degree of dispersion and adopt different strategies for different algorithm stages, including the feedback stage and the guidance stage.

Due to the lack of guiding force of dominant hippopotamus, we replace (15) with (16).

$$x_{i,j}^{\mathcal{H}ip\mathcal{E}} = \begin{cases} \mathcal{D}h + \tan(R\cdot\pi)\cdot\frac{ub_j^{cur} - lb_j^{cur}}{V_o} & \alpha_{Gui} > \frac{\mathcal{T}-t}{\mathcal{T}} \\ x_{i,j} + r_{12}\cdot\left(lb_j^{cur} + s\cdot\left(ub_j^{cur} - lb_j^{cur}\right)\right) & \text{else} \end{cases} \tag{16}$$

By utilizing St to store the individual that is the most recent dominant hippopotamus, the feedback result V_o can be obtained like this:

$$V_o = std(St) * \frac{200}{\left(ub_j^{cur} - lb_j^{cur}\right)}, \tag{17}$$

where $R \in [0,1]$, α_{Gui} determines how to guide the HO. St is referred to as learning experience. When hippopotamus of HO are dispersed, it is considered that HO is exploration, and when the historical hippopotamus are concentrated, it is considered that HO is exploitation.

Additionally, to prevent excessive guidance from inducing premature convergence of algorithm to local optima, a guidance threshold C_{Gui} is defined based on the Ebbinghaus memory curve [29] as

$$C_{Gui} = \left(1 - \frac{t}{\mathcal{T}}\right)exp(-\frac{(t-\alpha_{th})}{\mathcal{T}}) + \sum_{n=0}^{k}\frac{1}{\alpha_{th}}, \tag{18}$$

with k represents the total number of times the algorithm has been guided and α_{th} denotes the guidance strength, which is defined within the range $[1, \mathcal{T}]$. The threshold C_{Gui} incorporates a decay component to emulate forgetting and introduces a cumulative component to directly modulate the threshold for implementing guidance, thereby ensuring controlled guidance frequency while preventing excessive guidance.

To enrich the diversity of the hippopotamus population, we replace unstable random variables with disruption operator D_{op}, defined as

$$D_{op} = \begin{cases} Dis_{i,j} \times \psi(-0.5, 0.5) & Dis_{i,best} \ge \alpha_{Do} \\ 1 + Dis_{i,best} \times \psi\left(\frac{-10^{-16}}{2}, \frac{10^{-16}}{2}\right) & \text{else,} \end{cases} \tag{19}$$

where $Dis_{i,j}$ is the Euclidean distance between two hippopotamus, ith and jth (j is the nearest neighbor of the ith hippopotamus). Similarly, $Dis_{i,best}$ is the Euclidean distance between ith individual and $\mathcal{D}h$. $\psi(a,b)$ represents a random number uniformly distributed in the interval [a, b]. The parameter α_{Do} is set to 1 and is used to control the degree of disruption.

Disruption condition $C_{Do} = 1 - t/\mathcal{T}$ is a linearly decreasing function with time t.

$$Dis_{i,j}/Dis_{i,best} < C_{Do} \tag{20}$$

The disruption operator is set via (19) to ensure individual exclusivity, preventing insufficient search from the clustering of multiple high-fitness hippopotamuses. When C_{Do} meets the disruption condition (20), it shows the current hippopotamus is too close to $\mathcal{D}h$. Then, disruption operator D_{op} is constructed through (19), and used to update the hippopotamus's position via (21) and (22), with $\beta \in [0, 0.5]$.

$$x_{i,j}^{\mathcal{FB}hip} = \begin{cases} x_{i,j} + \beta\cdot\left(\mathcal{D}h - D_{op}\cdot\mathcal{M}_i\right) & T > 0.6 \\ t/\mathcal{T}\cdot x_{i,j} + \left(\frac{\mathcal{T}-t}{\mathcal{T}}\right)\left(\mathcal{D}h - D_{op}\cdot\mathcal{M}_i\right) & else \end{cases} \tag{21}$$

$$x_{i,j}^{\mathcal{M}hip} = t/\mathcal{T}\cdot x_{i,j} + (1 - t/\mathcal{T})\cdot x_{i,j}\cdot D_{op} \tag{22}$$

The disruption mechanism comprises two distinct components: a positional information term derived from candidate solutions that exhibits a t-dependent incremental progression, and a disruption factor demonstrating linearly as time t progresses. During the exploration phase, the candidate solution experiences controlled perturbation, which broadens the exploration of the parameter space and helps escape local optima. In the later stages, the candidate solution relies mainly on its current positional information.

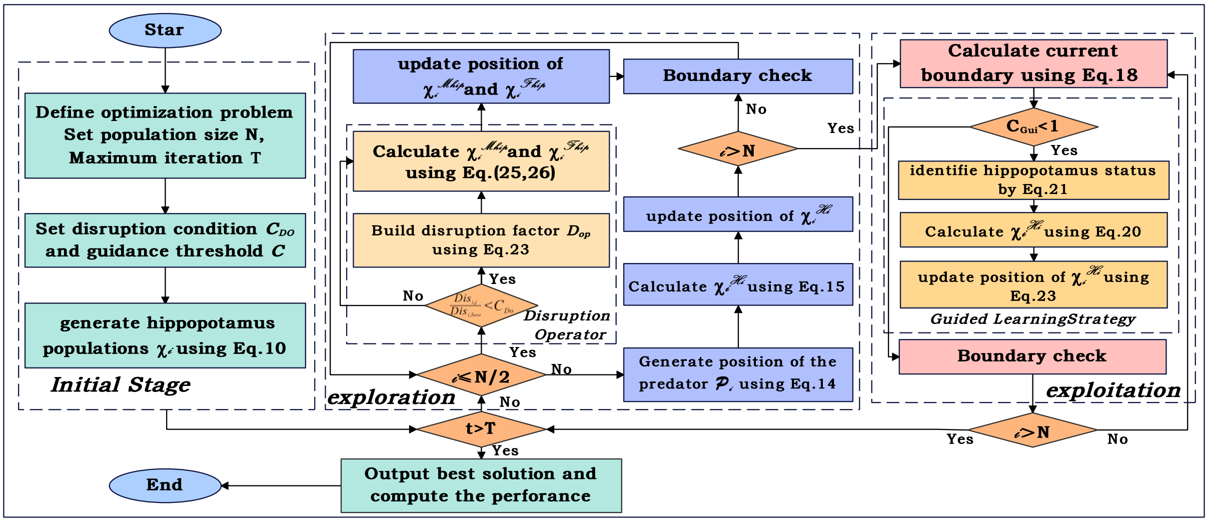


Fig. 1. Flow chart of the Guided-Improved Hippopotamus Optimization.

After completing population initialization, the GIHO algorithm transitions into its iterative optimization process. During the exploration phase, to resolve the issue of significantly suppressed population diversity caused by densely clustered hippopotamus agents in early stages and mitigate the risk of convergence to local optima, DO is employed to determine agent positions. GLS is implemented to address the insufficient guidance of $\mathcal{D}h$ during algorithmic iterations. Through the synergistic integration of DO and GLS, GIHO effectively addresses the challenge of balancing exploration and exploitation in HO, achieving enhanced optimization outcomes by dynamically regulating the equilibrium between global search capability and local refinement precision. Therefore, the flowchart of our proposed GIHO algorithm can be depicted in Fig. 1.

3.2. GIHO-VMD

To achieve adaptive optimization of VMD parameters, this paper proposes a collaborative framework based on FE and GIHO. The core idea is to quantify the complexity of modal components using FE and employ it as objective function to drive parameter search, thereby avoiding the efficiency challenge caused by directly embedding FE gradients in traditional methods.

The optimization function is defined as follows, where the fitness value $\mathcal{F}$ can be described as the complexity of a single signal. A smaller fitness value indicates that the signal contains fewer extraneous components.

$$\mathcal{F} = \sum_{k=1}^{K_{\chi_i}} FuzzyEn[u_k(t), m]. \tag{23}$$

When under-decomposition occurs in VMD, the multi-component signals within certain IMFs lead to elevated entropy values. Conversely, over-decomposition generates redundant IMFs, which also increases the total entropy. The proposed method optimizes the decomposition process of VMD by evaluating FE of individual IMFs, thereby enhancing the interpretability of the decomposition results.

Upon completing the GIHO iterations, the optimized parameter set $\mathcal{D}h$ (the optimal mode number K_{Dh} and penalty factor α_{Dh}) is implemented in VMD framework. These parameters, derived from the global FE minimization criterion.

4. Experimental setup and results analysis

4.1. Experimental condition and evaluation criteria

In this section, we evaluate the algorithm's performance in LS separation through qualitative waveform analysis and quantitative metrics, including Normalized Correlation Coefficient (NCC) and Signal-to-noise ratio (SNR), and their standard deviations. For simplicity, we term the proposed method M1. Many different current popular methods, including Grey Wolf Optimizer-based VMD method (M3) Li et al. [15], Polar Lights Optimizer-based method (M4) Yuan et al. [30], Black-winged kite algorithm-based VMD method (M5) Yang et al. [31], Whale Optimization Algorithm-based VMD method (M6) Wang et al. [32] are used to compare with our proposed approach. In order to reveal the improvement of GIHO, the HO-based VMD method (M2) is also chosen to compare with M1. To ensure the comparison is convincing and reliable, we allow each algorithm the same maximum iterations.

The Fundamental parameters setting of each algorithms are as follows. The size of population $\mathcal{N}$ is set as 30, number of decision variables m is set as 2, upper boundary and lower boundary of decision variables α, K are set as $[100, 8000]$ and $[3, 8]$, respectively, maximum number of iterations $\mathcal{T}$ is set as 100. The embedding dimension for fuzzy entropy is set to 2.

The experimental datasets are derived from public HS and LS datasets, ICBHI (International Conference on Biomedical Health Informatics) 2017 database [33] and PhysioNet/CinC Challenge 2016 [34]. For standardization, the signal duration is set to 5 s with a sampling rate of 2000 Hz. We randomly selected 100 HS samples and 100 samples of each of four LS types (Wheeze, Stridor, Rhonchi, and normal LS). To simulate real conditions, the samples are generated as breath sound acquisition signals by adding HLS mixtures and white Gaussian noise to achieve an SNR of 20 dB.

Recognizing that using only white noise as interference may not sufficiently assess the effectiveness of separation methods, this experiment additionally introduces non-stationary multimodal signals for a more comprehensive quantitative evaluation. Since acquired LS often contain noise with non-stationary characteristics, we refer to the approach proposed by K.D et al. [10] for evaluating VMD on such signals. Accordingly, a synthetic signal with intrawave frequency modulation is defined as

$$f_{noise}(t) = \frac{0.15}{1.8 + \cos(6\pi t)} + \frac{0.15\cos(64\pi t + 0.2\cos(256\pi t))}{2.5 + \sin(6\pi t)}. \tag{24}$$

As shown in Fig. 2, this synthetic signal with intrawave frequency modulation is used for our experiments. Therefore, nonlinear intrawave frequency modulation gives rise to a considerable number of higher-order harmonics. This characteristic clearly violates narrow-band principle fundamental to VMD, thereby complicating the recovery of this mode and raising the overall complexity of signal separation. The resulting waveform changes before and after mixing are shown in Fig. 3.

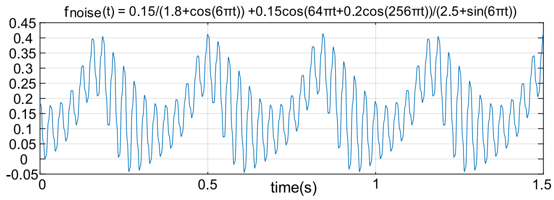


Fig. 2. A synthetic signal with intrawave frequency modulation used for our experiments.

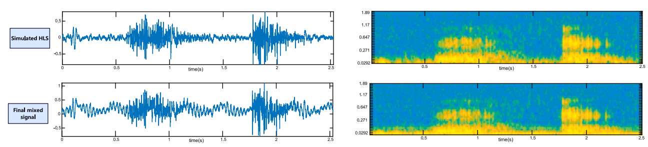


Fig. 3. Comparison of waveforms and Mel-spectrograms before and after introducing non-stationary noise in simulated HLS.

Experiments are implemented using the MATLAB (R2022A) on a computer system with Windows 10 (64 bits) and Intel i5-10200H CPU 16G RAM.

4.2. Qualitative experiment

We play back the original recordings and LS components separated by each method through a computer audio player, followed by auditory comparison. Based on the comparative results, M1, M2, and M4 are selected for further waveform and Mel spectrogram analysis due to their superior decomposition performance. Fig. 4 displays the separation results of M1, M2, and M4 for four distinct LS types.

The M1 demonstrates clearer advantages in pathological versus normal LS separation. In Fig. 4(a), the separated component M1 exhibits a continuous waveform devoid of high-frequency elements, consistent with normal lung sounds (LS) [35]. Compared to M2 and M4, which demonstrate notable energy attenuation, M1 presents uniform energy distribution in the Mel spectrogram without high-frequency concentration, along with smooth and continuous low-frequency energy. Furthermore, the inspiratory (breathing in) phase shows higher energy intensity and broader frequency bandwidth than the expiratory (breathing out) phase, aligning with criteria for pulmonary acoustics.

The separated component M1 in Fig. 4(b) manifests as continuous low-frequency sounds generated by airway secretion vibrations. Waveform analysis reveals that M1 preserves high-frequency components without the loss observed in M2 and M4 separation results. Its Mel spectrogram displays characteristic features of Rhonchi: concentrated low-frequency energy with intermittent bright patches in the low-frequency region, corresponding to transient airway wall vibrations induced by sudden mucus displacement [36]. In contrast, M2 and M4 demonstrate modal aliasing artifacts where cardiac acoustic components appear as interfering bright spots.

In Fig. 4(c), the separated waveform of M1 reveals periodic oscillations caused by fixed-frequency mucosal vibrations from airflow through constricted airways [37]. Unlike M2 and M4 that exhibit component loss, M1 maintains high similarity to the original signal. The Mel spectrogram of M1 decomposition contains wheeze-specific bright high-energy narrowband components with curvilinear striations, whereas these distinctive features undergo amplitude attenuation in M2 and M4 separations, resulting in indistinct narrowband representations. Fig. 4(d) demonstrates that the separated waveform of M1 presents high-frequency periodic patterns resembling fine serrations, characteristic of airway vibrations producing Stridor. The Mel spectrogram exhibits extremely narrow and concentrated high-energy bands within the 1000–2000 Hz range [38]. Compared to M2 and M4,

Table 1
The average optimization time achieved by M1 to M6 for separating simulated HLS at 60, 80, and 100 iterations.

Optimization	Optimization time (s)					
iterations	M1	M2	M3	M4	M5	M6
60	170.20	185.33	240.19	205.64	225.88	180.41
80	231.77	230.53	345.21	270.30	290.13	245.69
100	275.82	264.61	412.88	312.23	323.77	291.76

M1 demonstrates superior preservation of low-frequency stability and high-frequency energy retention characteristics.

Overall, M1 effectively isolates LS signals, with distinct waveform characteristics. The intensity of LS signals varies with LS types. The separated signals show a significant reduction in HS and noise interference, with waveforms clearly displaying defining features of LS.

4.3. Quantitative comparison experiment

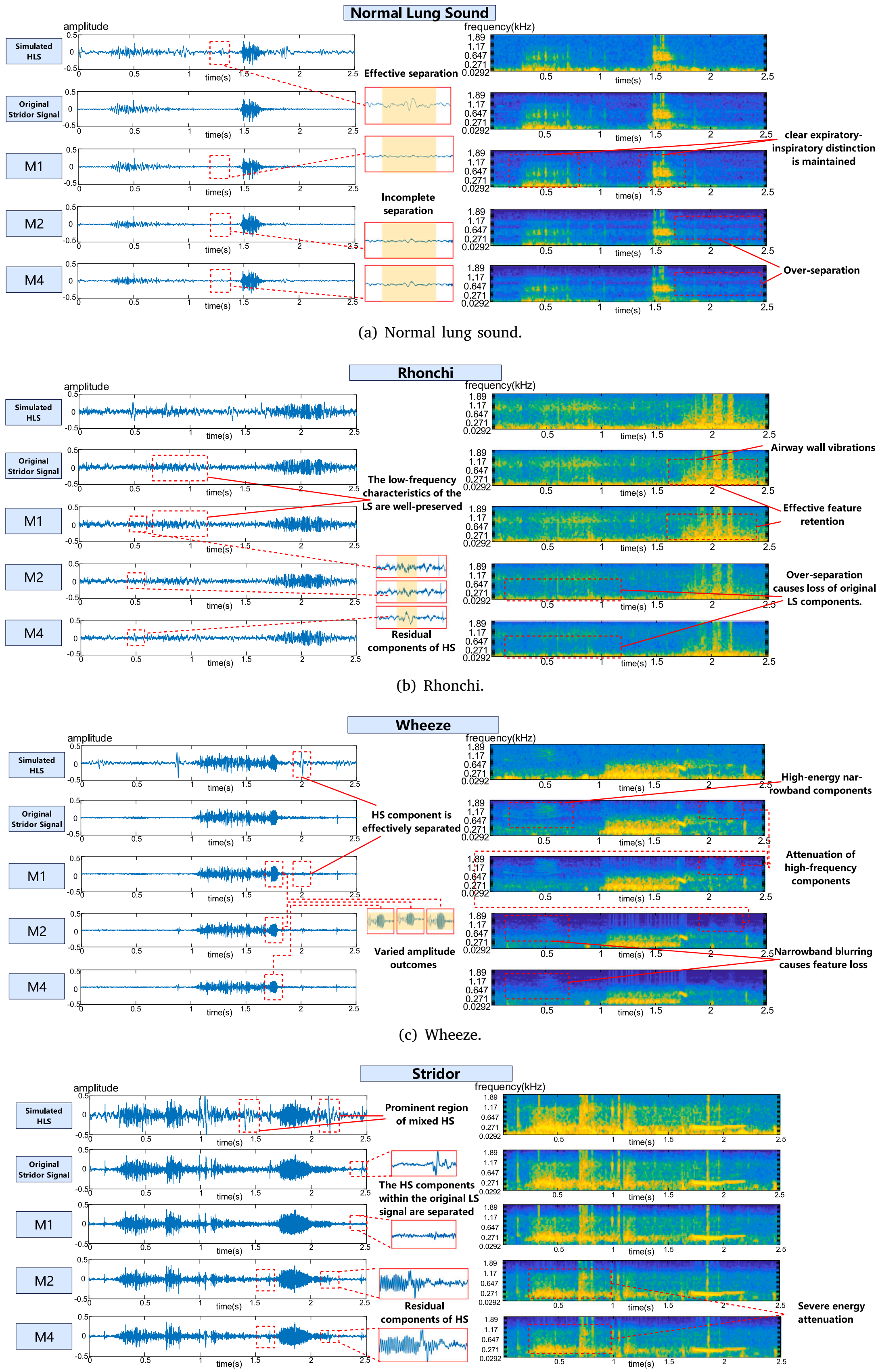
The separation performance is also quantitatively evaluated using SNR. As shown in Fig. 5, the proposed method M1 achieves a mean SNR of 30.27 dB for normal LS mixtures significantly outperforming other methods. For abnormal LS containing wheezes or rhonchi, M1 maintains SNR levels of 29.71 dB, with minimal impact from aliasing. In comparison, M2 yields relatively poor separation results with a mean SNR of 27.47 dB. The separation effectiveness of M4 and M6 only reaches an average SNR of 27.64 dB. The M3, lacking an effective disturbance mechanism, barely achieves an SNR of 27.4 dB at best. Even the best-performing M5 only achieves an average separation SNR of 28.3 dB.

In Fig. 6, the STD of separation results for each algorithm reveals that M2 and M4 has low stability, with STD value of 0.71 or higher, particularly when separating Wheeze and Stridor. Additionally, the STD for M5 and M5 separation results reaches 0.7 or higher. In contrast, M1 shows superior stability in separation efficiency, with an average STD of approximately 0.66 for the SNR of separated signals. The low STD value of indicates that it is less affected by variations in different types of LS due to its robustness against signal variations. The high stability of M1 is attributed to the GLS, which dynamically adjusts update mechanism of superior solutions based on current state of candidate solutions. Furthermore, the introduction of a dynamic threshold further enhances balance between exploration and exploitation.

The NCC values in Fig. 7 intuitively demonstrate that the LS components separated from HLS using M1 have higher similarity to original LS. M1 demonstrates superior performance in LS signal separation. For normal LS mixtures, the NCC between separated components and reference signals reaches 62.8% (mean), and abnormal LS mixtures also achieve NCC values of 53.55%–57.38%. Under identical iteration counts, conventional algorithms (M2-M6) yield significantly lower NCC ranges (36.72%–45.32%). To match M1's NCC performance, these baselines require 60–100 iterations.

The average optimization time required by methods M1 to M6 across different iteration counts is shown in Table 1. The test set comprises all simulated HLS mixtures, and all optimization procedures are implemented in MATLAB (R2022a). Each method is run on the same experimental platform, as previously described. Results indicate that M1 consumes less time than other methods under the same number of iterations. In contrast, both M3 and M6 exhibit considerably longer computational times.

Furthermore, the separation results for the test HLS signals remixed with non-stationary multimodal noise (as shown in Fig. 3) are presented in Table 2. The incorporation of such noise partially simulates a realistic noisy environment, which consequently affects the separation outcomes of all methods. Nevertheless, as shown in the table, M1 still demonstrates outstanding separation capability along with high computational efficiency. It achieves separation results with SNR exceeding



(a) Normal lung sound.

(b) Rhonchi.

(c) Wheeze.

(d) Stridor.

Fig. 4. Comparison of waveforms and Mel-spectrograms before and after denoising for various LS.

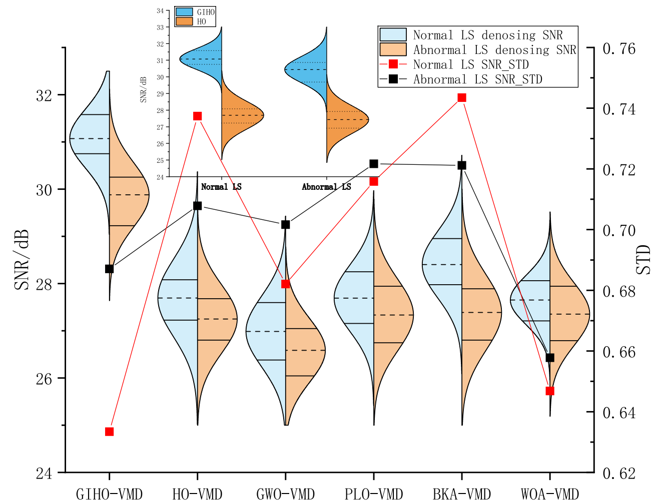


Fig. 5. SNR values obtained from separating 20 sets of LS signals (each set includes 3 normal and 3 abnormal hybrid HLS segments) using different algorithms.

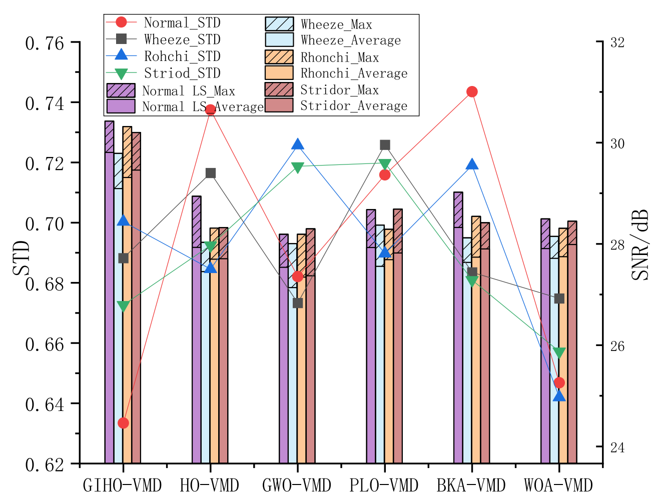


Fig. 6. The SNR of denoising results for four types of mixed heart-lung sound using six algorithm models.

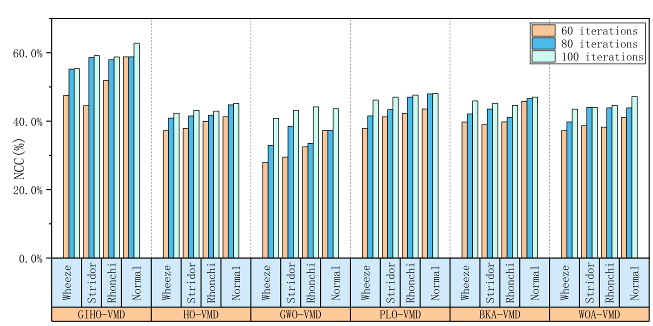


Fig. 7. The average NCC value of the denoising results of various algorithms in 60, 80, 100 iterations.

28.3 dB and NCC exceeding 51% on average for all four types of mixed lung sounds. while also requiring the shortest optimization time among all methods.

The comparison validate M1's effectiveness in retaining critical LS features while achieving physiologically plausible separation. This enhancement stems from introduction of DO, which allows GIHO to disruption parameter combinations with high FE. By preventing premature convergence through diversity preservation, thereby ensuring the global search capability of GIHO.

In conclusion, the GIHO-VMD method demonstrates statistically significant improvements in LS separation. It achieves 12.3–13.8% higher SNR and 15.2% lower STD than existing metaheuristic-VMD algorithms, while also improving computational time efficiency by approximately 18%–24% compared to current optimization methods. Furthermore, it achieves an NCC exceeding 62.8% for normal LS cases. Therefore, GIHO-VMD is more suitable for LS separation than other currently algorithms.

5. Discussion and conclusion

Although VMD-based approaches have demonstrated significant efficacy in LS separation, most existing methods rely solely on frequency-domain features [39,40]. Moreover, their separation performance is critically constrained by the sensitivity to hyperparameter settings. Consequently, when processing pathological HS-LS mixtures with spectral overlap, the separation results are frequently compromised by either over-decomposition, leading to critical LS component loss, or under-decomposition, causing persistent HS residuals. To address these issues, we propose GIHO-VMD. This approach employs both frequency and series complexity as dual-scale decomposition criteria. Specifically, the GLS in GIHO-VMD eliminates spectral contamination in separated LS caused by suboptimal parameter selection, while the DO mitigates premature convergence to local optima, thereby preventing spectral leakage in the decomposed components. Separation experiments conducted on synthetic HLS — generated from two public databases (ICBHI-2017 and CinC 2016) — demonstrate that the proposed method achieves statistically superior separation accuracy compared to existing algorithms. The BKA and PLO schemes exhibit high waveform similarity for normal LS; however, variability in separation stability is empirically observed across different LS types due to suboptimal disruption strategies during the exploitation phase. While consistent stability is preserved by WOA and GWO schemes across diverse LS types, insufficient optimization causes persistent residual noise contamination in decomposed components, resulting in degraded lung sound purity. In contrast, the proposed approach delivers separated LS with optimal NCC and SNR, while demonstrating consistent separation results and uniform stability across diverse pathological variants.

Although our method demonstrates robust performance in lung sound signal separation, it exhibits several limitations. First, the incorporation of signal sequence complexity as a metric employs an oversimplified approach, leading to insufficient balance between frequency and complexity during decomposition. Moreover, the optimization model entails computationally intensive operations, requiring significant improvements in time efficiency. Finally, it should be noted that the experimental results are based solely on simulated mixtures of lung and heart sounds. While this approach allows for controlled initial validation, it constitutes a clear limitation. The performance and generalizability of the method on real clinical auscultation data remain unverified and warrant further investigation. To address these limitations, future work should prioritize the following directions:

(1) Designing a multi-scale decomposition method. Combining VMD with FE reduces excessive dependence on frequency characteristics in decomposition results. However, limiting FE solely to the objective function significantly weakens its regulatory effect during VMD iterations. This constraint consequently hinders effective control over the complexity of individual IMFs.

(2) Streamlining iterative logic in optimization algorithms. Compared to state-of-the-art methods, the GIHO algorithm enhances update strategies to achieve superior optimization performance. However, balancing exploration and exploitation phases requires complex computational procedures, which remains a key limitation constraining GIHO's computational efficiency.

Table 2
The performance of M1 and the other five methods in separating various types of LS mixed with non-stationary noise.

Method	Normal LS			Rhonchi			Wheeze			Stridor		
	SNR/dB	NCC	Optimization time/s	SNR/dB	NCC	Optimization time/s	SNR/dB	NCC	Optimization time/s	SNR/dB	NCC	Optimization time/s
M1	29.61	62.53%	271.88 s	28.97	58.13%	288.51 s	28.37	51.84%	281.84 s	29.14	58.75%	278.61 s
M2	26.17	47.65%	263.18 s	27.48	43.55%	291.51 s	27.04	43.18%	313.49 s	27.64	41.47%	283.67 s
M3	26.23	46.91%	367.84 s	26.43	44.17%	388.64 s	26.14	42.19%	375.42 s	27.55	45.10%	376.47 s
M4	27.20	49.37%	318.28 s	27.18	48.87%	361.49 s	26.93	46.75%	348.27 s	28.61	47.64%	352.48 s
M5	27.83	49.18%	315.81 s	27.81	47.79%	346.88 s	26.84	47.03%	337.38 s	28.37	46.27%	328.84 s
M6	27.13	51.13%	322.53 s	28.13	48.10%	388.43 s	27.54	46.07%	371.46 s	28.34	47.43%	355.78 s

(3) Evaluating the generalizability of lung sound separation methods in real-world clinical settings remains essential. Although the proposed approach demonstrates robust performance on simulated datasets, authentic clinical auscultation involves multi-position signal acquisition where the realistic distribution of lung sounds, heart sounds, and environmental noise cannot be fully replicated in simulation. Therefore, future work should focus on developing comprehensively annotated datasets that capture auscultatory signals from standardized thoracic locations. Such datasets would enable rigorous validation of separation performance using clinically representative data. Furthermore, integration with deep learning methods [41] should be explored to achieve tasks with practical clinical significance, such as classification, identification, and prediction.

CRediT authorship contribution statement

Leiming Zhang: Writing – original draft, Methodology, Formal analysis. **Fuliang He:** Writing – review & editing, Supervision, Conceptualization. **Hao Tan:** Resources, Data curation. **Wangquan Wang:** Data curation. **Shiyuan Wang:** Supervision.

Ethics statement

The data used in this study were publicly available data sets on the Internet. No animals or humans were victims.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix

To address the constrained problem presented in Eq. (2), a Lagrangian augmented with a quadratic penalty term is introduced, which can be formulated as

$$\mathcal{L}\left(\{u_k\},\{\omega_k\},\lambda\right)=\alpha\sum_k\left\|\partial_t\left[\left(\delta(t)+\frac{j}{\pi t}\right)*u_k(t)\right]exp(-j\omega_k t)\right\|_2^2 + \left\|f(t)-\sum_k u_k(t)\right\|_2^2+\left\langle\lambda(t),f(t)-\sum_k u_k(t)\right\rangle, \tag{25}$$

we first express the representation of as given in Eq. (26), where each u_k and ω_k is taken as the latest available update.

$$u_k^{n+1}=\underset{u_k\in X}{\arg\min}\left\{\alpha\left\|\partial_t\left[\left(\delta(t)+\frac{j}{\pi t}\right)*u_k(t)\right]e^{-j\omega_k t}\right\|_2^2 + \left\|f(t)-\sum_i u_i(t)+\frac{\lambda(t)}{2}\right\|_2^2\right\} \tag{26}$$

By applying the Parseval–Plancherel Fourier isometry under the norm $\mathcal{L}^2$, this optimization problem can be transformed and efficiently solved in the spectral domain.

$$\hat{u}_k^{n+1}=\underset{\hat{u}_k,u_k\in X}{\arg\min}\left\{\alpha\left\|j\omega\left[\left(1+\mathrm{sgn}(\omega+\omega_k)\right)\hat{u}_k(\omega+\omega_k)\right]\right\|_2^2 + \left\|\hat{f}(\omega)-\sum_i\hat{u}_i(\omega)+\frac{\hat{\lambda}(\omega)}{2}\right\|_2^2\right\}. \tag{27}$$

We now perform a change of variables in the first term, and by exploiting the Hermitian symmetry of the real signals within the reconstruction fidelity term, both terms can be rewritten as half-space integrals over non-negative frequencies.

$$\hat{u}_k^{n+1}=\underset{\hat{u}_k,u_k\in X}{\arg\min}\left\{\int_0^\infty 4\alpha(\omega-\omega_k)^2\left|\hat{u}_k(\omega)\right|^2 +2\left|\hat{f}(\omega)-\sum_i\hat{u}_i(\omega)+\frac{\hat{\lambda}(\omega)}{2}\right|^2 d\omega\right\}. \tag{28}$$

The solution for the mode u_k and the center frequencies in quadratic optimization problem are shown as

$$\hat{u}_k^{n+1}(\omega)\leftarrow\frac{\hat{f}(\omega)-\sum_{i<k}\hat{u}_i^{n+1}(\omega)-\sum_{i>k}\hat{u}_i^{n}(\omega)+\frac{\hat{\lambda}^n(\omega)}{2}}{1+2\alpha\left(\omega-\omega_k^n\right)^2}. \tag{29}$$

Since the center frequencies ω_k are only present in the bandwidth prior and not in the reconstruction fidelity term, the problem simplifies to:

$$\omega_k^{n+1}=\underset{\omega_k}{\arg\min}\left\{\left\|\partial_t\left[\left(\delta(t)+\frac{j}{\pi t}\right)*u_k(t)\right]e^{-j\omega_k t}\right\|_2^2\right\}. \tag{30}$$

The optimization is carried out in the Fourier domain, whereby the quadratic problem is readily solved as

$$\omega_k^{n+1}\leftarrow\frac{\int_0^\infty\omega\left|u_k^{n+1}(\omega)\right|^2 d\omega}{\int_0^\infty\left|u_k^{n+1}(\omega)\right|^2 d\omega}. \tag{31}$$

Data availability

The experiments described in the article all use public datasets and have obtained the consent of the authors.

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